

Assignment 8 - Phonology

Ling 201 – Spring 2026

(100 pts total)

Please **do not use external sources such as the internet or AI to find the answers**. If you need an **extension**, please ask before the deadline.

Part I. Identifying Minimal Pairs

Are the following pairs of English words minimal pairs or not? If so, what phonemic contrast do they illustrate (give the IPA symbols for the two sounds that contrast)? If not, explain briefly in your own words why they are not a minimal pair. Remember to focus on sounds, not spelling. **Do NOT count a near-minimal pair as a minimal pair.** (4 points/item, 20 points total)

1. peel – pill

minimal pair for [i] – [ɪ]

2. song – strong

not a minimal pair, they differ in more than one sound

3. flirt – dirt

not a minimal pair, they are a near-minimal pair

4. write – right

not a minimal pair, they have exactly the same pronunciation (homophones)

5. altitude – aptitude

minimal pair for [l] and [p]

Part II. Providing Minimal Pairs

For each pair of contrasting phonemes, give a minimal pair (**NOT a near-minimal pair**) of **English words** that illustrates the phonemic contrast (write your words in standard English spelling, not the IPA). (4 points/item, 20 points total) [multiple answers are possible]

1. /k/ – /t/

cap - tap

2. /ʃ/ – /tʃ/

ship - chip

3. /ʌ/ – /æ/

but - bat

4. /f/ – /θ/

fin - thin

5. /ɪ/ – /ɛ/

bid - bed

Part III. Phonological Processes

Consider how the pronunciation of the following words has changed in the development of Spanish. For each word, **indicate** each individual sound segment that has changed. For each sound segment that has changed, **state** which phonological process has affected the sound and **briefly explain why that phonological process was involved**. Choose from insertion, deletion, assimilation, dissimilation, weakening, strengthening. (4 points/item, 20 points total)

1. Latin *legale* ‘legal’

→

Spanish *leal* ‘loyal’

[g]-deletion

[e]-deletion

2. Latin *spata* → Spanish *espada* ‘sword’

[e]-insertion

[t] to [d] voice assimilation to the voice of neighbouring vowels / [t] to [d] weakening

3. Latin *mensa* → Spanish *mesa* ‘table’

[n]-deletion

4. Spanish *tan bien* ‘as well’ → *también* ‘also’ (ignore the accent mark)

[n] to [m] assimilation to the place of neighbouring [b]

5. Latin *nomine* → *nomne* → *nomre* → Spanish *nombre* ‘name’

[i]-deletion

[n] to [r] dissimilation of manner (nasal) from neighbouring [m]

[b]-insertion of the same place with neighbouring [m]

Part IV. Phonemic Analysis: Korean

Consider the following data from Korean, a language isolate spoken in Korea. Pay attention to the sounds [s] and [ʃ] and then answer the questions below based on this data. For each word, the pronunciation is given in brackets and the meaning is given in quotation marks.

- | | | | |
|------------------|----------------|--------------|---------------|
| 1. [ʃi] | ‘poem’ | 10. [sal] | ‘flesh’ |
| 2. [miʃin] | ‘superstition’ | 11. [kasu] | ‘singer’ |
| 3. [ʃinmun] | ‘newspaper’ | 12. [sanmun] | ‘prose’ |
| 4. [tʰaksanʃige] | ‘table clock’ | 13. [kasəl] | ‘hypothesis’ |
| 5. [ʃilsu] | ‘mistake’ | 14. [miso] | ‘smile’ |
| 6. [oʃip] | ‘fifty’ | 15. [susek] | ‘search’ |
| 7. [panʃik] | ‘method’ | 16. [tapsa] | ‘exploration’ |
| 8. [kanʃik] | ‘snack’ | 17. [so] | ‘cow’ |
| 9. [kafi] | ‘thorn’ | | |

Answer the following questions based on the dataset above. (points in total)

1. List all of the phonetic environments in which [s] and [ʃ] occur. (6 points)

[s]	[ʃ]
k_a	#_i
l_u	i_i
#_a	#_i
a_u	ŋ_i
#_a	#_i
a_ə	o_i
i_o	ŋ_i
#_u	n_i
u_e	a_i
p_a	
#_o	

2. Do [s] and [ʃ] occur in the same or different phonetic environments in this dataset? What does this tell us about the distribution of these two sounds? Are they two separate phonemes or allophones of the same phoneme? (6 points)

These two sounds never appear in the same environments according to question 1, which means they have fallen in complementary distribution. They are allophones of the same phoneme thereby.

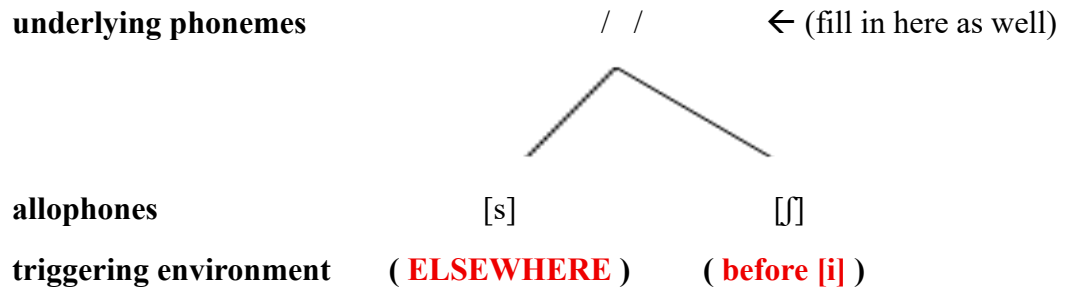
3. List left and right environments (no repetition) for [s] and [ʃ]. (8 points)

left environments		right environments	
[s]	[ʃ]	[s]	[ʃ]
{k, l, #, a, i, u, p}	{#, i, ŋ, o, n, a}	{a, u, ə, o, e}	{i}

4. Which side of environment decides the sound change? Why? (4 points)

right environments, there is no overlap between the two sounds' right environments

5. Fill in the diagram to decide the underlying phoneme. One of the triggering environments should be “ELSEWHERE” (4 points)



6. Use your own words to describe the diagram, recall what we have done in class for arctic foxes. (4 points)

underlying /s/ is realised by default as [s] sound, unless it is before [i] sound, where it will be realised as [ʃ] sound.

7. Write down a phonological rule based on question 6. (4 points)

/s/ → [ʃ] / _ [i]

8. Suppose we have two words as below (they might not exist in actual Korean language). Finish up the table. (4 points)

phonemic form	/asasa/	/asasi/	underlying representation
rule: /s/ → [ʃ] / _ [i]	-	asaʃi	intermediate representation
phonetic form	[asasa]	[asaʃi]	surface representation

(Don't forget to fill in the rule.)

Extra Credit

This part is **optional**. If you attempt it, you will receive *at least 1 extra point* on top of your grade. Up to **5 points** may be awarded based on the quality of your answer. (5 points)

Consider the following phrases in English and read them out loud. Are you able to relate them to the Korean puzzle above in Part IV? Don't worry, since this question aims to convey an idea that different languages might have the same phonological processes. Therefore, be creative!

[HINT: [j] sound and [i] sound are very similar phonetically.]

1. I miss **you**!
2. Nice to meet **you**!
3. Bless **you**!
4. Guess **you** are right.
5. Unless **you** forgot to bring your homework, you are fine.
6. Did **you**?
7. Would **you** like some help?
8. Put **your** hands up!
9. I wish I were as smart as **you**.
10. Is **your** answer correct?

Certain consonants in English change when they occur before the sound /j/ (the “y” sound): /s/ becomes [ʃ] (*miss you* → [mɪʃu]), /z/ becomes [ʒ] (*is your* → [ɪʒɔr]), /t/ becomes [tʃ] (*meet you* → [mi:tʃu]), and /d/ becomes [dʒ] (*did you* → [dɪdʒu]). Because /j/ is phonetically similar to the vowel [i] (both are front/palatal sounds), this is parallel to the Korean pattern where /s/ becomes [ʃ] before [i]. In both languages, sounds change in a predictable way depending on their phonetic environment, especially before front or palatal sounds.